Research status & verification

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Rank-eight multiplicity-three continuation and current verification scope

Read these results and data with caution.
AI-generated research · independent verification pending. The results, proofs, code and data have not yet been independently verified by human experts or formalized in Lean. Use them with caution; exact checks establish only their stated computational scope.

The current classification record

The rank-eight multiplicity-three continuation addresses all 3,527 supplied exact-multiplicity-three rings: 3,506 general exclusions, 16 models and 5 unresolved cases. Every recorded model admits a unitary realization. The remaining rings retain unresolved status for general complex categorification. The rank-seven multiplicity-five continuation addresses all 5,805 supplied exact-multiplicity-five rings: 5,804 general exclusions, 1 realization and 0 unresolved cases. The classification of the supplied census is complete through multiplicity 5. The new realization R7_5_5678 is Rep(AGL(1,7)). The two earlier rank-six questions remain open. The historical 38-case inventory keeps its original scope.

Rank	Bound	Rings	Excluded	Models	Open
4	128	889,530	889,515	15	0
5	20	40,999	40,982	17	0
6	9	25,421	25,362	57	2
7	5	10,006	9,976	30	0
8	3	4,500	4,402	93	5

Ranks two, three and four are classified without a multiplicity restriction. Exactly fifteen rank-four Grothendieck rings occur, and every rank-four categorification admits a spherical structure. The table records the separate finite censuses.

The bound is the largest multiplicity included in the displayed census, not necessarily a complete classification range. Rank six is complete through multiplicity three; its exact bands 5, 6, 7 and 9 are also complete, while bands 4 and 8 each contain one open case.

Rank-six models comprise 56 unitary constructions and one spherical construction that is not pseudo-unitary. Completeness means coverage of the stated census by the project's recorded arguments; it does not assert independent human or formal verification.

The rank-seven M5 tensor identities, fusion axioms and preservation of the earlier tensor set passed independent computational checks. Completeness of the exhaustive enumeration relies on the supplied producer completion record; the optimized enumerator was not independently rerun.

The rank-eight M3 extension preserves all 973 earlier records and adds 3,527 exact-multiplicity-three tensors. Source identities, tensor axioms and based-isomorphism audits are independently checked. Enumeration completeness relies on producer completion records. The supplied optimized source matches the producer source hashes for both the rank-seven M5 and rank-eight M3 runs; the exhaustive enumerations were not independently rerun.

MATHEMATICAL FRONTIER

7 open cases

Each question concerns general complex fusion categorification. The reductions below are necessary conditions with explicit limits.

R6_4_155 / Odd cyclic Clifford constraints

Every odd generalized pivotal rotation must factor through a cyclic spin character. All 1,144 tested blocks pass. The all-order path-Clifford divisibility condition for the source family holds exactly at even parameter values, including the parameter four of this case.

The unique rank-26 modular candidate and rank-22 based-module witness remain necessary data. No compatible associators, coherent module category or general obstruction is established.

R6_8_1660 / Complete indicators and central phase masses

In the positive spherical representative, saturated modular representations determine all indicator sequences and every central phase mass. The eight spin objects form four dual pairs, all with restriction 2X+2gX+sigma+3T.

Exact phase Gram constraints leave 640 labelled restriction-data profiles in three branches of 68, 286 and 286. These profiles are neither categories nor complete modular fusion data. They do not construct compatible associators or exclude all categorifications.

Rank eight: 5 rings / Exact multiplicity three

The new band contains 3,506 recorded general exclusions, 16 unitary models and 5 unresolved tensors. Each remaining tensor has an individual attempt record, distinguishing completed necessary-condition tests, unsuccessful model searches, ineligible tests and time limits.

These are investigated categorification candidates, not asserted fusion categories or established exotic examples. Passing necessary conditions supplies no associator. Failing to find a model does not prove nonexistence. Exact tensors are in data/remaining_cases.py; the per-case record is evidence/rank8_creative_20261022/per_ring_research_audit.json.

These remaining rings are extensively investigated exotic candidates in the operational sense: each survived the documented case-specific campaign of known and newly developed exclusions and model searches, including complete finite domains and specified unbounded construction families. This designation does not assert existence or novelty of a future realization.

In the stated finite center systems, R8_3_928 has four exact integer-phase witnesses and six pattern exclusions certified by complete branching trees with 49 exact Farkas leaves; R8_3_941 retains 14 patterns, with 11 integer witnesses and 3 open integer-feasibility questions in its earlier Gram test. These witnesses are necessary data, not categorical models; uncertified numerical failures are not exclusions.

R8_3_721 is realized by the inherited 1+rho Q-system Morita dual of the C2 de-equivariantization of GH(C8). Exact rank and coefficient bounds identify its unique dimension type in the supplied complete rank-eight multiplicity-at-most-three census. The earlier center-action analysis leaves three necessary R8_3_1154 patterns. These results are in evidence/rank8_next_20261021/; the later R8_3_65 general exclusion is in evidence/rank8_research_20261022/.

New structural criteria

E21 unitary realization and creative rank-eight restrictions

R8_3_982 is unitarily realized by the neutral triality component of the exceptional E21 chiral fusion category from SU(3)_21 into E7_1. All 512 canonical fusion coefficients are identified exactly; the construction realizes both nonregular based-module types and the algebra 1+X5. Every Morita partner of any R982 categorification has the same based fusion ring. Five rank-eight general categorification questions remain: R8_3_928, R8_3_941, R8_3_1154, R8_3_1768 and R8_3_3472. Every hypothetical categorification admits a spherical structure and has a pseudo-unitary Galois conjugate; none has a braided Morita representative. New saturated-conductor and signed spectral-capacity theorems constrain all twist orders. R1768 has 39 of 40 necessary Gram/indicator branches excluded; its irrational dimension-ratio sector outside the two dimension-orbit objects has 18 or 72 simples. The final cone survives, and the 18-object option has a forced 19-dimensional Weil block with integral nonnegative distinguished products. R1154 has four forced semionic S columns and fusion matrices in all 25 histograms, which remain necessary data. R3472 has seven remaining first-action dual spectra and 400477 second-action arithmetic spectra, conditional on the specified rank-two modules. Exact integer phase counts eliminate six of the ten R928 center patterns, leaving four with exact necessary witnesses; all fourteen R941 allocations survive the stated phase refinements. Scoped construction obstructions and such witnesses neither exclude these five rings nor construct exotic categories.

A generalized Haagerup Morita construction and further restrictions

R8_3_721 is realized by the inherited 1+rho Q-system Morita dual of the C2 de-equivariantization of GH(C8). Exact rank and coefficient bounds identify its unique dimension type in the supplied complete rank-eight multiplicity-at-most-three census. The earlier center-action analysis leaves three necessary R8_3_1154 patterns. These results are in evidence/rank8_next_20261021/; the later R8_3_65 general exclusion is in evidence/rank8_research_20261022/.

A uniform vertex-line obstruction proves that these inherited cyclic generalized Haagerup Q-system duals have trivial invertible group. Complete free-bimodule Gram factorization, the known center and rational coefficient inequalities then identify the C8 example. This is an existing algebra in an existing category; modular data alone are not used to assume a categorification.

The historical R8_3_982 center search had a surviving semionic branch; its unitary E21 realization is established in the later creative stage. The charged fourth-moment systems for R8_3_928 and R8_3_941 retain exact feasible necessary completions.

Additional center-pattern refinement

Fourth Verlinde moments couple all distinguished tensor-product decompositions to the same nonnegative integer restriction matrix. Exact minimum-norm projection and rank-one residual Grams yield coordinate obstructions. A complete cyclic-space and orthogonal-projector argument classifies the remaining distinguished center-action tuples for R8_3_721.

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Further rank-eight frontier investigation

The original 34 unresolved M3 tensors now have 25 additional general-complex exclusions. Their identifiers are R8_3_436, R8_3_439, R8_3_455, R8_3_567, R8_3_731, R8_3_901, R8_3_903, R8_3_921, R8_3_925, R8_3_926, R8_3_985, R8_3_1055, R8_3_1203, R8_3_1350, R8_3_1463, R8_3_1615, R8_3_1617, R8_3_1660, R8_3_1661, R8_3_1739, R8_3_2842, R8_3_2885, R8_3_3258, R8_3_3344, R8_3_3413.

The new arguments include simultaneous split-module representations, normal-matrix spectral support, a complete near-group TPE reduction, complete central-induction frame constraints, full character-labeled modular S and STS equations, a two-prime norm criterion for integral center objects, complete integer center decompositions and cubic Verlinde integrality, and integral Morita and relative-cohomology obstructions. Exact all-parameter filters exclude specified near-group construction routes for every original frontier tensor. General Galois-pseudo-unitarity reductions extend the scope of all seven noncommutative searches. Complete connected-algebra underlying-object filters and exact feasible center constraints describe the remaining construction possibilities; their survivors are not asserted algebras or categories.

Rank-eight multiplicity-three continuation

Unified exclusions combine negative-subring transfer, quadratic-compositum compression, weak-integral and biquadratic relative twist traces, complete signed lifting and convex central induction. Exact separation of an enlarged necessary-condition system gives a valid contradiction only after the hypotheses and complete lifting are established.

Structural arguments use module-category ranks, split-module polynomial constraints, solvability, equivariantization and Morita transport. A unitary-only positivity obstruction becomes a general exclusion only when a separate theorem proves the relevant category must admit a unitary realization.

The explicit models include near-group equivariantizations, a cyclic permutation equivariantization of three Fibonacci factors, exceptional subfactor constructions, a conformal-embedding Morita dual, ordinary groups and a twisted A6/A5 group-theoretical construction. Direct model comparisons check all 512 coefficients; the new GH(C8) Morita dual is identified by an exact coefficient bound and uniqueness in the supplied complete census. Existence does not imply uniqueness or categorical novelty.

Rank-seven multiplicity-five continuation

The new continuation combines quadratic Casimir compression, complete signed sphericalization, Galois reduction to positive dimensions, and exact simultaneous first and second center-twist moments. A separate integral case is excluded by a complete cyclic-module Morita-dual calculation and descent to dimension 148. The current manuscript gives the categorical reductions and the finite domains checked by the programs.

The center-trace certificates impose no arbitrary cutoff on root orders. The conductor-sixty extension pairs relative-trace values under modular Galois symmetry; it does not assume equality of individual paired twists. Structural reductions and unsuccessful model searches alone do not change a case to excluded.

A general equality criterion applies at any prime p congruent to one modulo four: a twist-fixed quadratic-torus vector with saturated compression eigenvalue plus or minus 1/sqrt(p) generates a representation of level p, and its nontrivial twist support lies in the corresponding Legendre class of primitive pth roots. The manuscript proves this at every prime-power level and applies p=5 to the quadratic pair.

A second uniform criterion treats the odd conductor-eight character. Its partial projections give exact mass ratios for two-primary twist orders one, two, four and eight. Duality and Galois actions fixing the complete source field then yield two finite alternatives for active twist orders. Both criteria are proved at all congruence levels and supply necessary conditions for the complete finite-certificate arguments.

R7_5_5039 is now excluded in the general complex scope by coupled fourth and eighth rotations and an integral fusion-norm inequality. An all-order relative-phase inventory covers arbitrary twist orders. Two conductor-eight branches have exact modular contradictions; the complementary branches reduce exhaustively to 48 integer trees with 96 exact separating leaves. The eighth indicators force the finite packet domain, while the final separators use fourth-rotation and fusion-integrality constraints. This is a complete source-bound argument, distinct from rejecting an individual necessary-data witness.

R7_5_3895 is also excluded in the general complex scope. Complete signed and Galois reductions, both low induction frames, every second-indicator alternative and both character labellings are covered. The final coupled-rotation systems have 240 fourth-rotation choices: 225 have exact rational contradictions, and the remaining fifteen have complete integer chains with 2,126 exact leaves. A finite integral-order calculation checks all low words without a degree cutoff and proves the necessary norm-slack parity. The

complete source-bound verifier passes in normal and optimized Python with SciPy unavailable, including a repeated run in one process.

Rank eight: an inner-square obstruction

For R8_2_24, a hypothetical complex categorification has a pseudo-unitary Galois conjugate with a rank-four neutral component. Saturated modular compression forces its second indicators. Conjugation by an odd invertible would fix every neutral simple label and square to conjugation by the neutral involution.

A simultaneous pivotal chart and the short weights of associator coefficients show that such a square root would yield infinitely many inequivalent tensor functors between fixed fusion categories. Tensor-functor rigidity excludes this, giving a general complex obstruction. The source and finite-weight checks support the argument within their stated scope; its categorical reasoning remains subject to independent mathematical verification.

> **Status discipline.** A restricted nonexistence result applies only under its stated assumptions. An exact computation of conditional equations does not discharge the missing categorical hypothesis.

EXACT COMPUTATIONAL EVIDENCE

What was checked

E21 unitary realization and creative rank-eight restrictions

The E21 and creative stage has a separate exact replay and accepted source manifest. Run python3 tools/verify_rank8_creative_20261022.py and python3 -O tools/verify_rank8_creative_20261022.py. Its proof and verifier inventory is evidence/rank8_creative_20261022/README.md. The --historical option replays prior rank-eight mathematics from its immutable archive.

These targeted checks supplement the original M3 and 108-check frontier replays; those historical receipts do not certify the later stage. Metadata-only checks reconcile source identities without repeating mathematics.

A generalized Haagerup Morita construction and further restrictions

The new construction has separate normal and optimized checks, including independent reconstruction of the coefficient inequalities and an exhaustive scan of all 4,500 supplied census entries. Its reviewed categorical proof and bound source files are retained in evidence/rank8_next_20261021/. The original evidence and 108-check replay receipts remain unchanged.

Run tools/rank8_next.py --check for the accepted overlay identities. The model proof lists its targeted arithmetic verifiers. Census completeness remains the supplied producer result; the full enumeration has not been independently rerun.

Additional center-pattern refinement

The separate center-pattern addendum passed all eight checks in canonical normal and optimized Python, including independent arithmetic reconstructions and positive controls. Fresh replay after extraction from the publication archive is deferred. Run evidence/rank8_center_followup_20261021/verify.py --root . in both modes.

This earlier refinement preserves the frozen frontier decision ledger and its original audit. It established necessary pattern reductions only; the subsequent GH(C8) construction is recorded separately.

Further rank-eight frontier investigation

Canonical normal and optimized frontier replays each passed all 108 checks. These receipts cover the historical frontier stage. Fresh replay after extraction from the publication archive is deferred.

The original M3 ledger and replay receipts are preserved. New decisions use a separate continuation ledger and source manifest. Run tools/verify_rank8_frontier.py in normal and optimized Python after the original M3 replay. Its strict metadata-only verification still requires both matching full continuation receipts.

The per-ring audit records completed finite exclusion tests, exact feasible necessary-condition witnesses and excluded construction families. Each result keeps its precise hypotheses and search domain.

Rank eight: multiplicity-three extension

Fresh full normal and optimized aggregate replays passed for the accepted source-bound ledger. Their receipts bind the exact source tensors, historical decisions, accepted certificates, model constructors and proof-source files. Each referenced replay log is separately checksummed.

Run supplements/rank8/software/verify_m3_continuation.py in normal and optimized Python with --output pointing to separate validation receipts, preserving the frozen baseline receipts. Assertion-based inherited checks run with assertions enabled. Metadata-only mode checks file identities and reconciliation with the full receipts; it does not repeat the mathematical computations. The release includes the repository-relative dependency tree.

Exact replay checks the stated finite identities and certificates. Categorical reductions, complete enumeration and claims of mathematical novelty have separate proof obligations. Independent human verification and Lean formalization remain pending.

Rank seven: multiplicity-five extension

The extension is kept separate from all frozen earlier proof stages. The preserved M4 ledger is restored for historical replay; current status partitions and reader downloads include the M5 records.

Exact normal and optimized aggregate replays passed for the current ledger.

Run supplements/rank7/software/verify_m5.py for the extension and consult its normal and optimized receipts under supplements/rank7/output/m5/. These checks certify the implemented input identities, criterion certificates and preserved historical decisions; they do not independently prove categorical coherence or exhaustive-enumeration completeness.

Ranks four and five

The unrestricted rank-four proof gives FPdim < 1000 and multiplicity at most 31 without assuming pivotality. Its exact finite classification checks 1,692 rings, 1,677 exclusions and 15 models. Every rank-four complex fusion category admits a spherical structure; the sharp bounds are FPdim $\leq 24+12\sqrt{3}$ and multiplicity ≤ 6. The theoretical arguments and exact computations have distinct verification obligations.

Both complete rank-four replays passed: 889,530 rings, 15 models and 889,515 exclusions. Each regenerated 6,470,006 signed orbits; an independent full unsigned recount also passed.

The reconstructed rank-five package passed all 40,999 pinned classes, 17 model matches and 40,982 general exclusions in normal and optimized Python. The two reconstructed ledgers agree byte-for-byte. The unavailable source ledger was not recovered; the reconstructed ledger has its own recorded digest.

Ranks seven and eight

The complete base-certificate loops passed in normal Python. Separate source-bound transition replays and current-ledger checks passed in normal and optimized Python with exact parent restoration. These checks account for the current partitions displayed in this note.

The rank-eight base replay checks all 973 records, including 75 signed proof trees, four rank-seven subring proof transfers and six appendix tensor transfers. The R8_2_24 rigidity package separately checks its canonical tensor, neutral fusion rules and finite associator-weight claims; the categorical proof has separate AI audits.

Rank-six necessary conditions

The odd cyclic Clifford checker passes all 1,144 tested blocks and its invalid-input controls. Independent exact indicator and phase-mass checks cover both conjugate choices for the second residual; the complete profile allocation leaves 640 labelled possibilities. Normal and optimized replays pass within this necessary-condition scope.

Rank six: a combined verification campaign

The complete initial process checked all 25,362 exclusion certificates: 25,355 passed and seven failed. Each of those seven then passed the complete unchanged checker under SymPy's Python backend. Exact matrix comparison attributes the failures to equation ordering or orientation and positional separating-vector conventions; no proof inequality or certificate was relaxed.

Both post-loop checks skipped by the initial failure gate passed separately. The model registry gives 57 matches, and the multiplicity-one partition gives 39 = 18 + 21. A hash-bound aggregate validates complete certificate coverage and 299 source/input bindings.

This is a combined campaign, not a successful single full run or an all-Python full-census pass. The initial process exited with failure, and that status is preserved in the evidence. Its loaded backend/version was not measured at startup; the global python-flint installation changed from 0.8.0 to 0.9.0 during the run, so worker and child processes may have loaded different versions.

The complete-replay orchestrator has passed identity and rejection-control checks but has not itself been executed end to end. Its existence must not be reported as an additional full-census success.

READING AND REPRODUCING THE RECORD

Evidence and its limits

Source identity

A checksum binds a statement or calculation to specified bytes. It cannot establish that their mathematics is correct.

Exact replay

A program verifies its implemented finite equations, enumerations or tensor comparisons. Its mathematical assumptions remain part of the proof obligation.

Categorical argument

A claimed construction or obstruction must establish the hypotheses connecting the finite calculation to a fusion category. Source and arithmetic checks support a categorical proof only within their stated scope.

Independent review

Separate AI review passes are not independent human verification. No human verification or Lean formalization is claimed here.

Replay from the repository root

```
python3 tools/restore_data.py
python3 tools/verify_current_publication.py
python3 -O tools/verify_current_publication.py
```

Add --historical to run the complete source-bound certificate chain on isolated inputs. Checkers using assert require normal Python; an optimized invocation alone removes those assertions. The current-publication verifier also checks the identity and consistency of the published documents and their website copies.

The rank-specific full-census commands, dependencies, receipts and limitations are documented in the verification guide. The aggregate release checks are separate from the mathematical certificate replays.

Current sources

Research status
Verification guide
Public research status page

The manuscript statements, individual ledger entries and their source-bound evidence should be read together. This note summarizes the current scope; it is not an independent proof certificate.